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Boloorian et al.

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(54) **USE OF FREQUENCY OFFSETS IN GENERATION OF LIDAR DATA**

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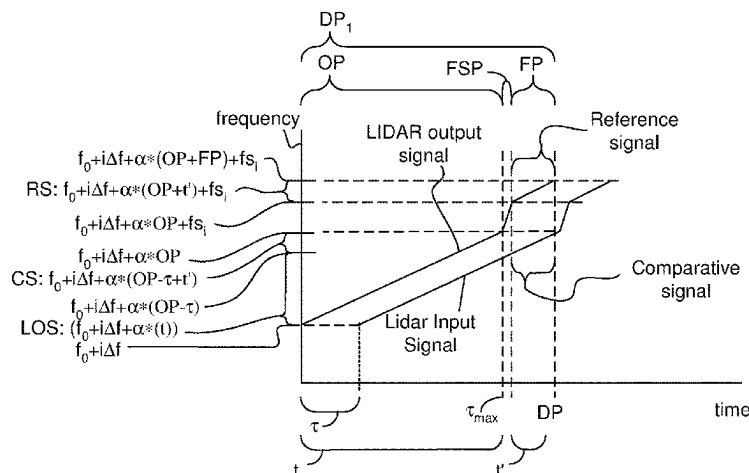
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(57) **ABSTRACT**

A LIDAR system includes a light source that outputs an outgoing LIDAR signal that includes multiple different channels. The LIDAR system also generate multiple composite light signals that each carries a signal couple and are each associated with a different one of the channels. A signal couple includes a reference signal and an associated comparative signal. The comparative signals each include light from the outgoing LIDAR signal that has been reflected by one or more objects located outside of the LIDAR system. The reference signals also include light from the outgoing LIDAR signal but also exclude light that has been reflected by any object located outside of the LIDAR system. There is a frequency differential between a frequency of the reference signal and a frequency of the associated comparative signal. The frequency differential includes a contribution from a frequency offset that is induced by electronics. The electronics induce the frequency offset such that the frequency offset is different for each signal couple.

8 Claims, 10 Drawing Sheets



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See application file for complete search history.

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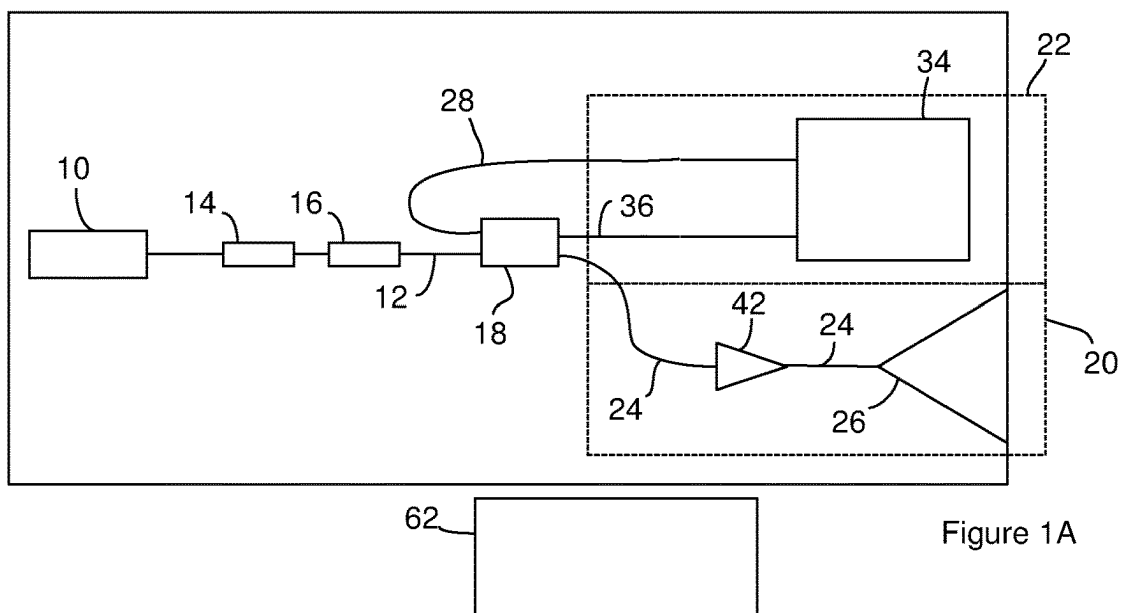


Figure 1A

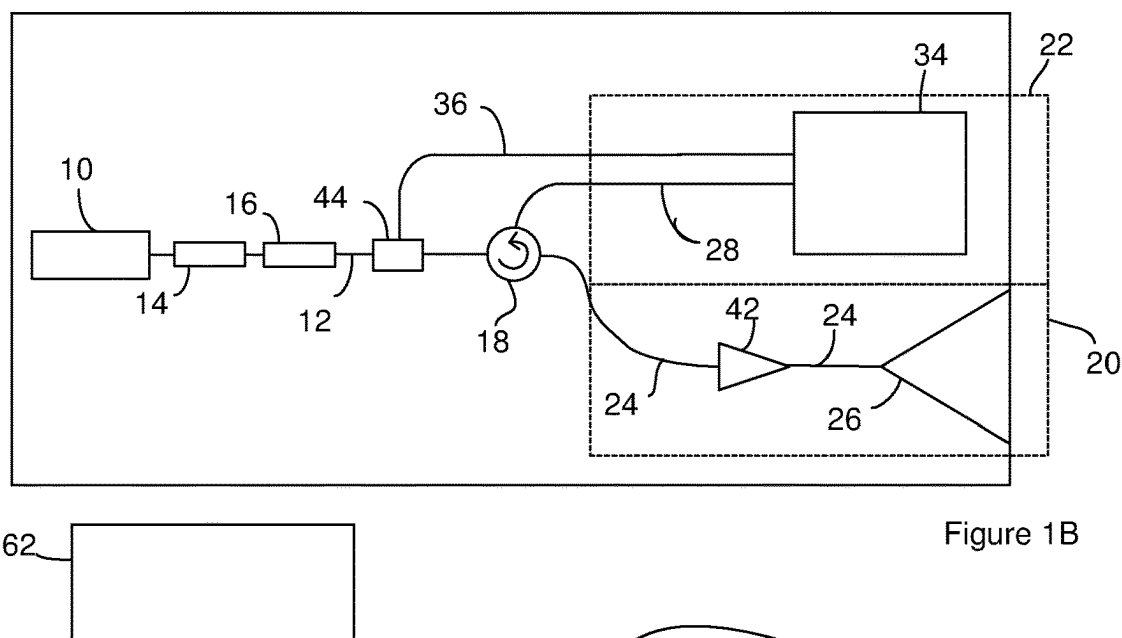


Figure 1B

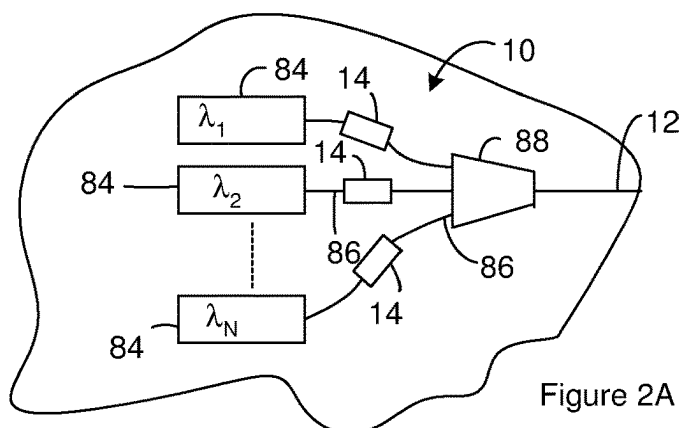


Figure 2A

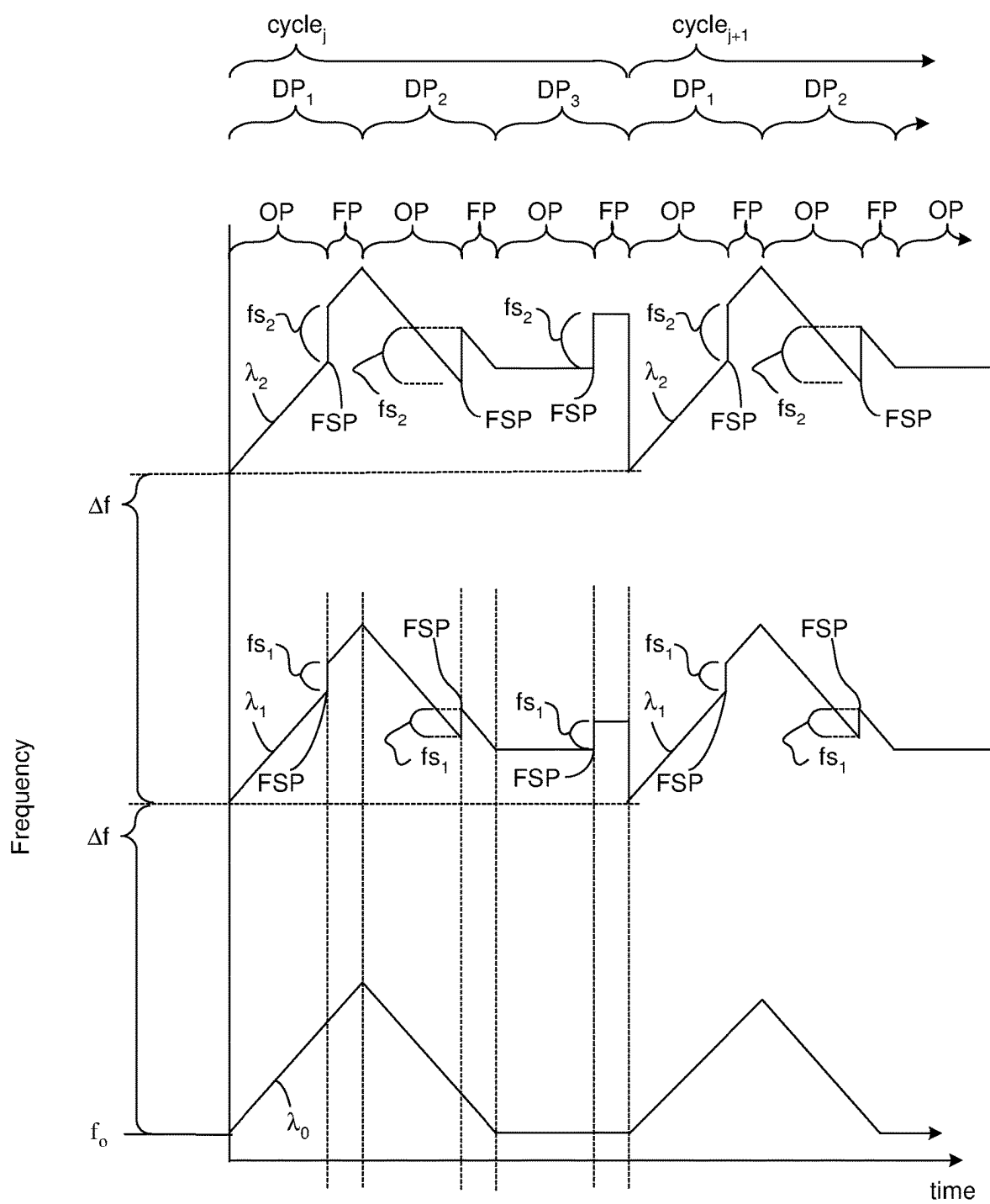


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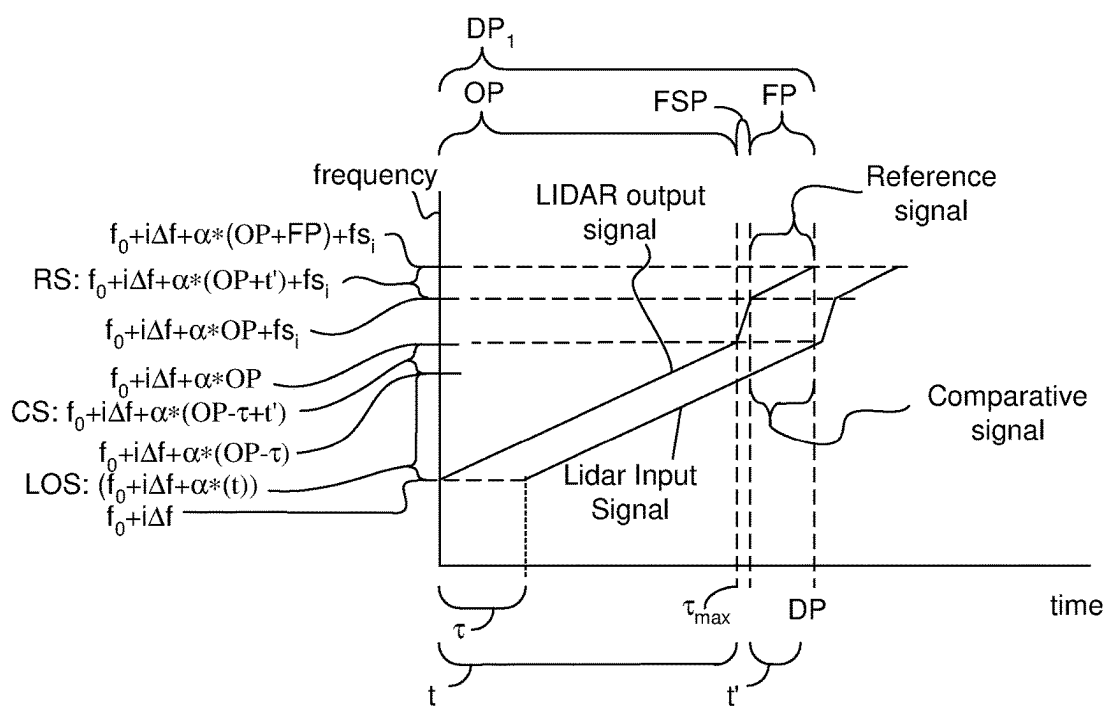


Figure 2C

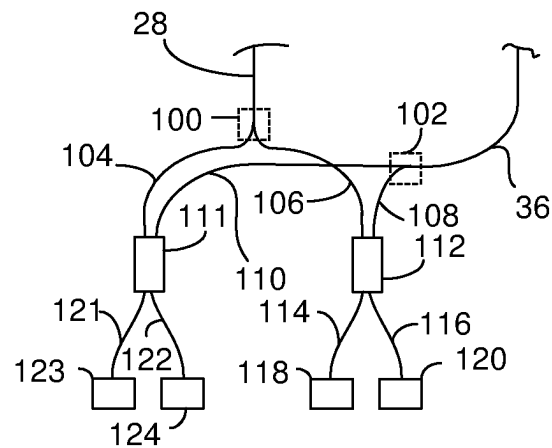


Figure 3A

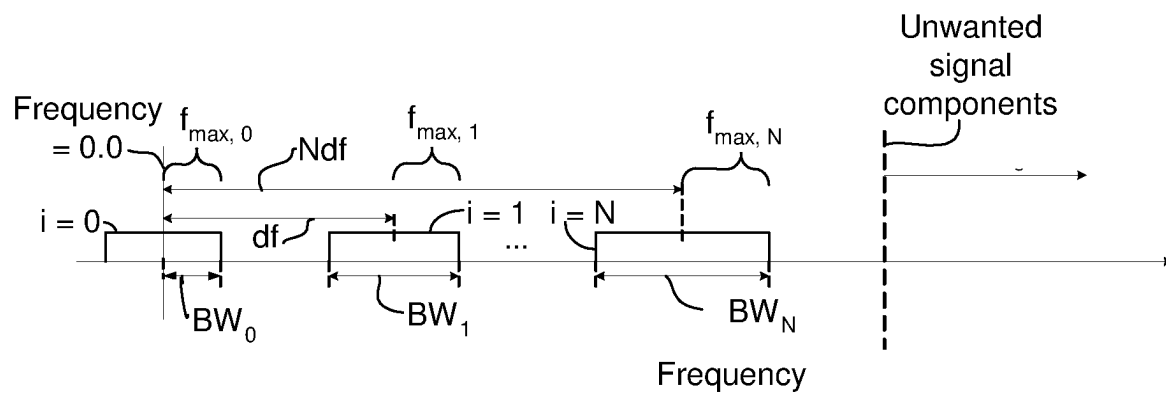


Figure 3B

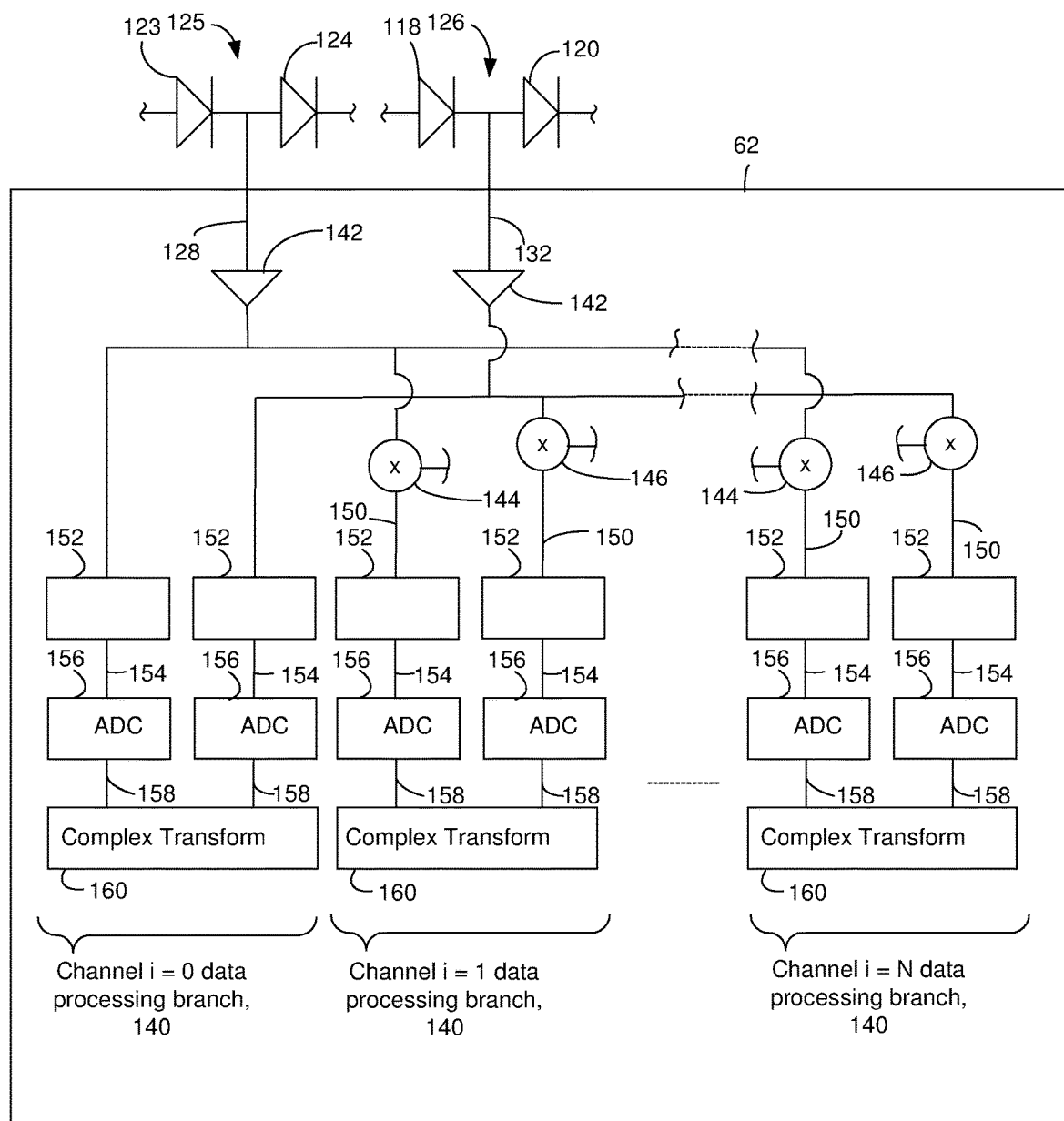


Figure 3C

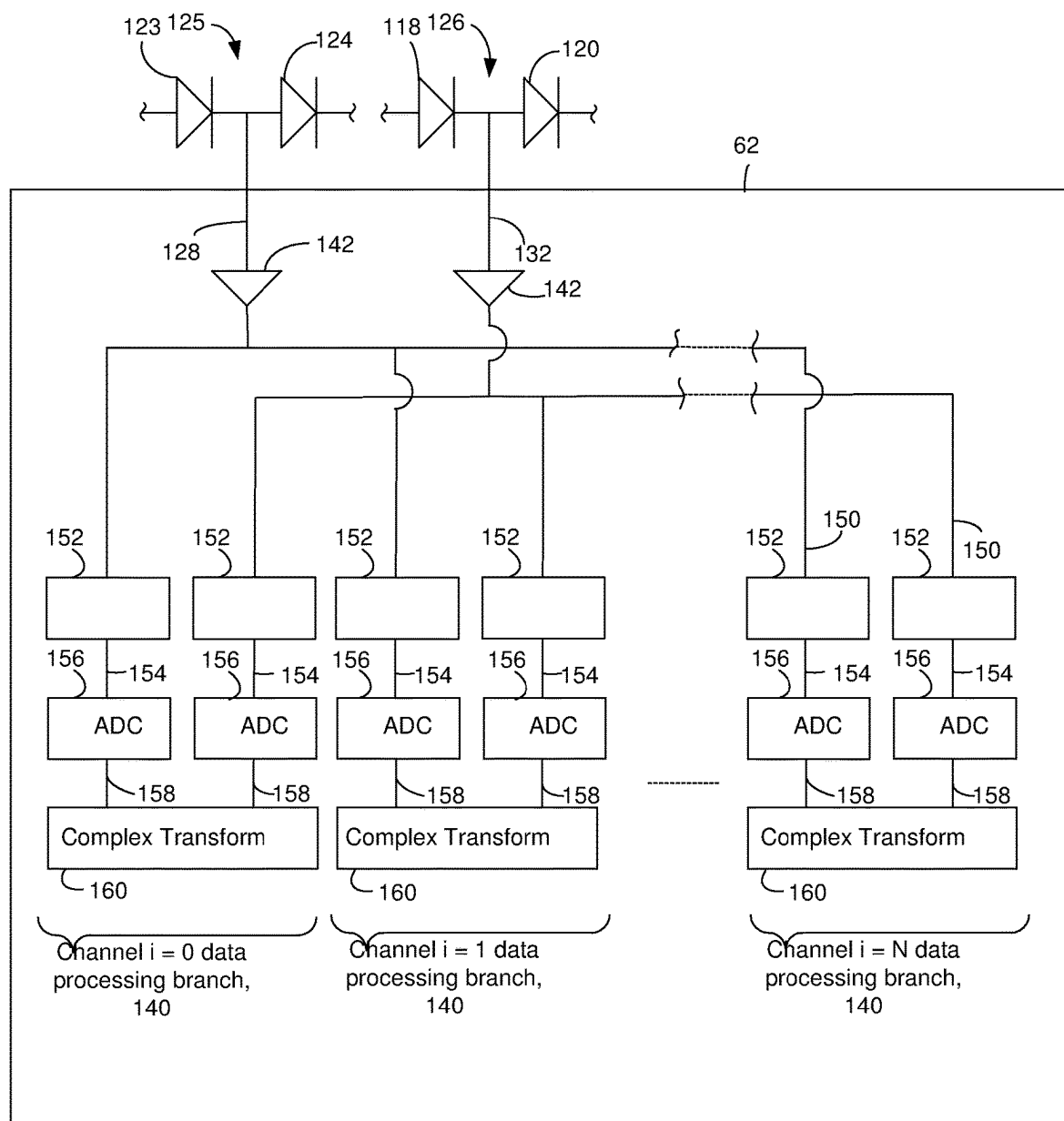


Figure 3D

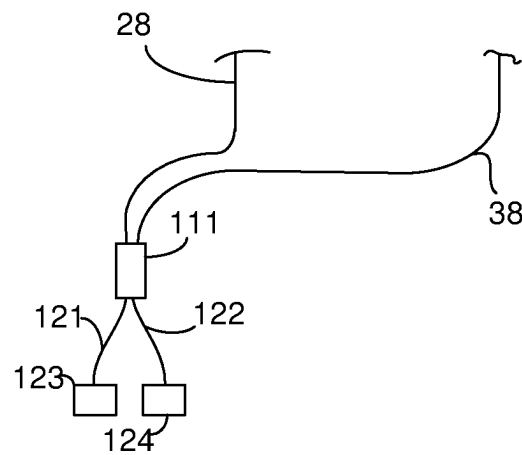


Figure 4A

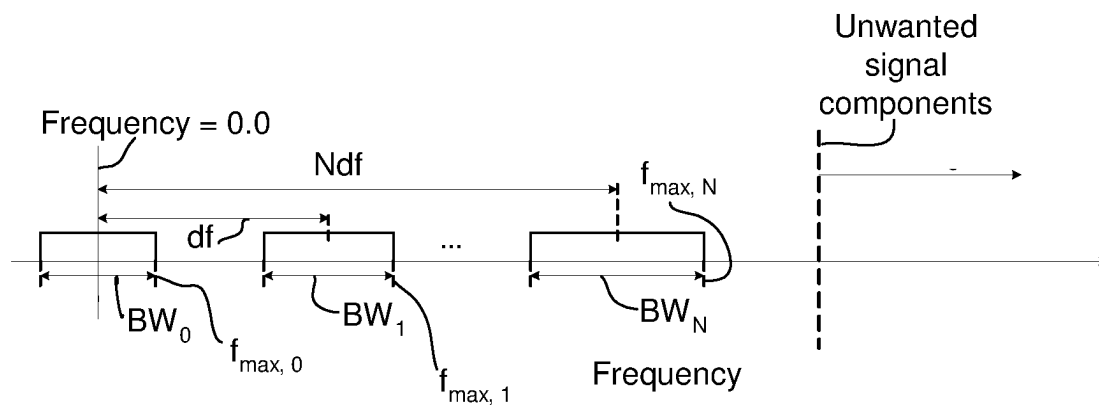


Figure 4B

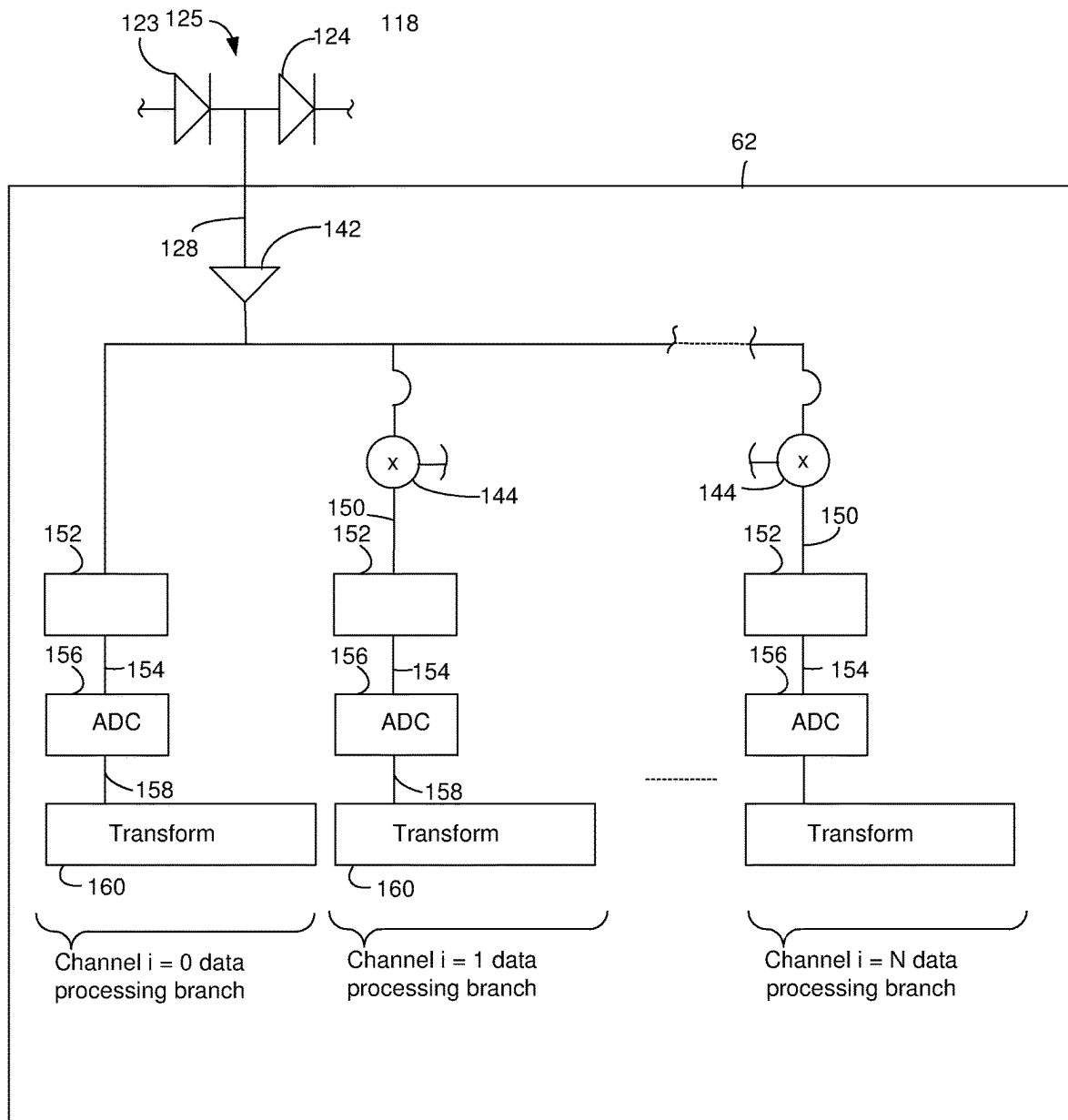


Figure 4C

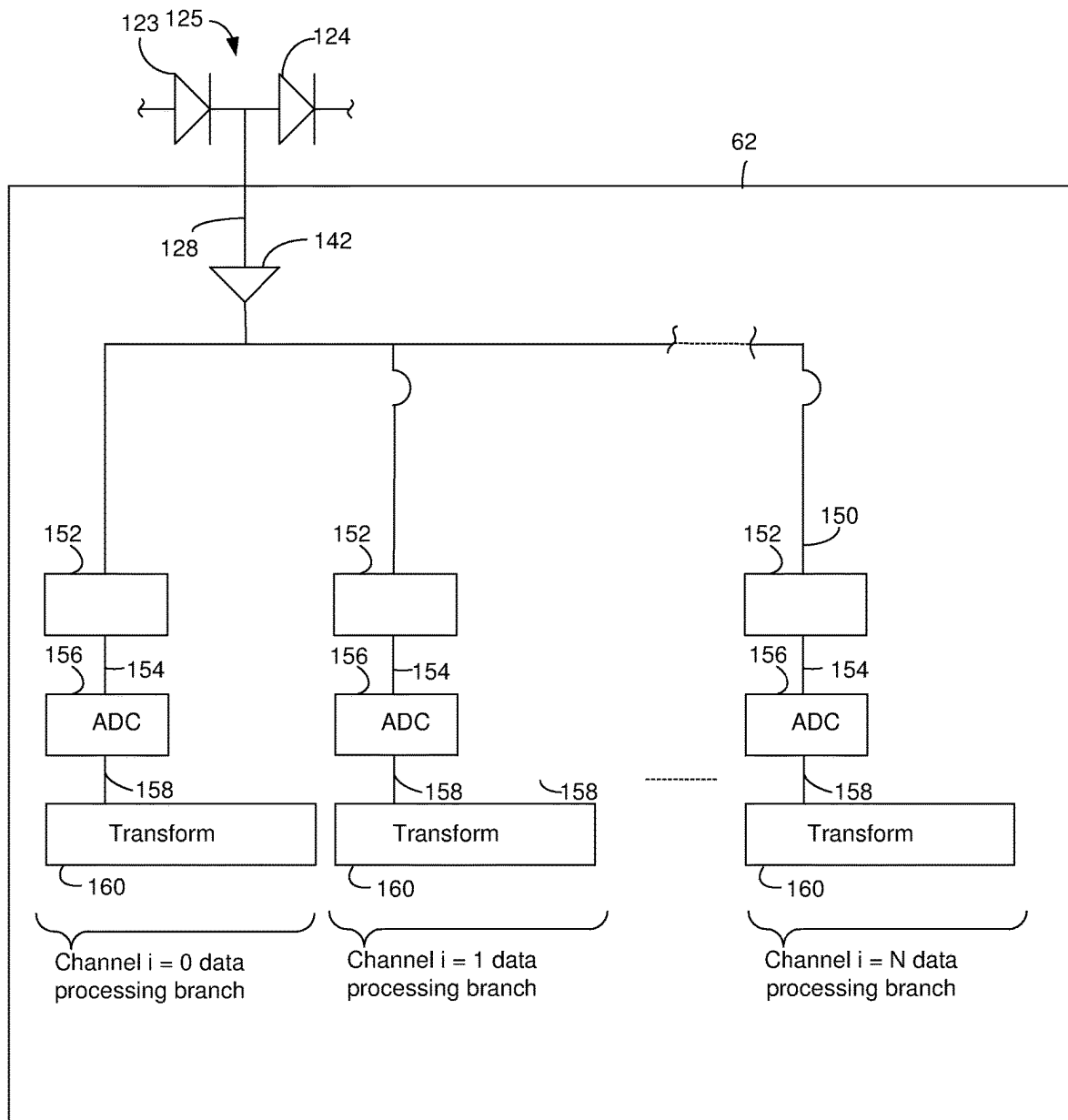


Figure 4D

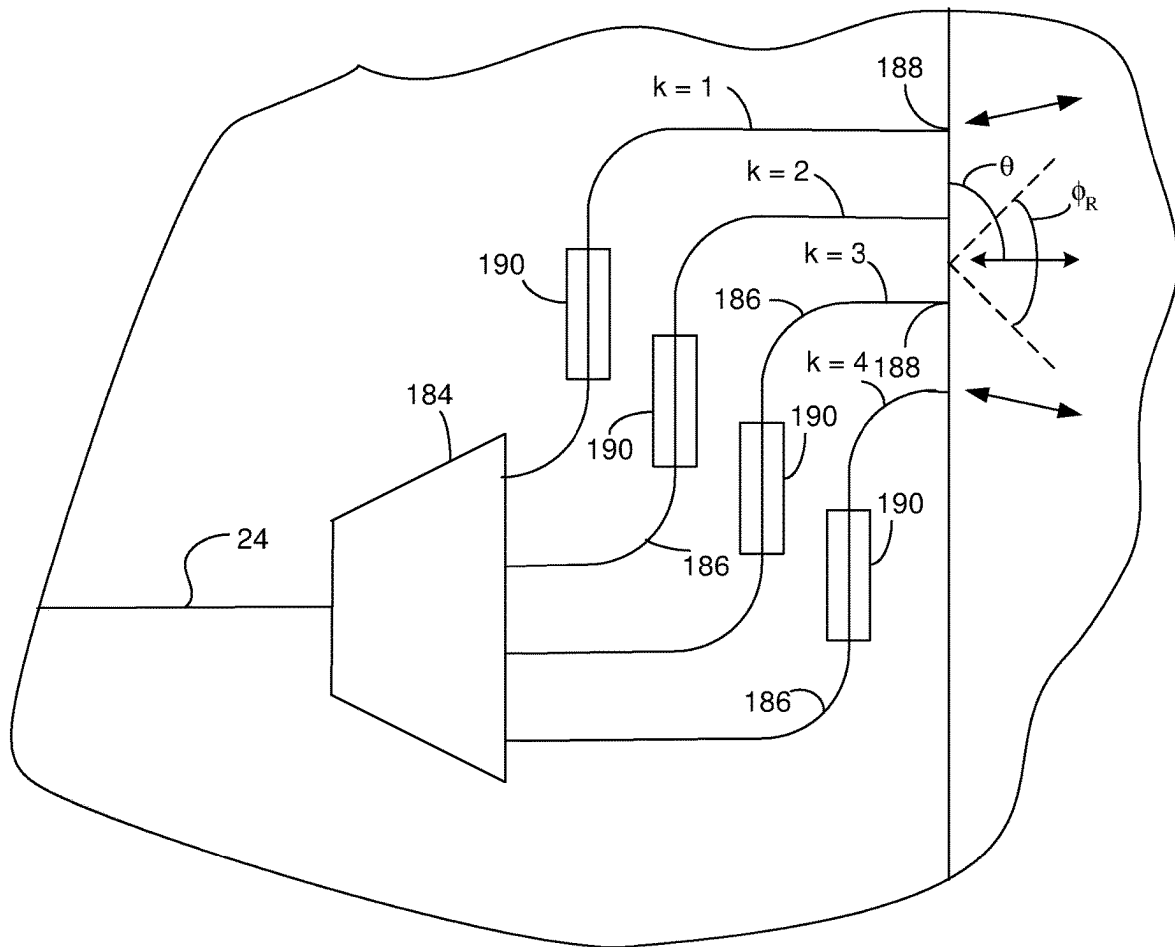


Figure 5

1

USE OF FREQUENCY OFFSETS IN GENERATION OF LIDAR DATA

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/542,648; filed on Aug. 16, 2019; entitled "Use of Frequency Offsets in Generation of LIDAR Data;" and incorporated herein in its entirety; and U.S. patent application Ser. No. 16/542,648 is a continuation-in-part of U.S. patent application Ser. No. 16/457,443; filed on Jun. 28, 2019; entitled "Use of Frequency Offsets in Generation of LIDAR Data;" and incorporated herein in its entirety.

FIELD

The invention relates to optical devices. In particular, the invention relates to LIDAR systems.

BACKGROUND

LIDAR technologies are being applied to a variety of applications. LIDAR specifications typically specify that LIDAR data be generated for a minimum number of sample regions in a field of view. LIDAR specifications also specify the distance of those sample regions from the LIDAR signal source and a re-fresh rate. The re-fresh rate is the frequency at which the LIDAR data is generated for all of the sample regions in the field of view. The ability of the given LIDAR system to generate the LIDAR data for the sample regions in the field of view becomes more difficult as the distance to the sample regions increases and as the refresh rate increases.

As LIDAR is being adapted to applications such as self-driving-vehicles, it becomes more desirable to generate LIDAR data for larger fields of view, increasing numbers of points, further distances, and at faster re-fresh rates. As a result, there is a need for a LIDAR system that capable of generating LIDAR data for larger numbers of sample regions.

SUMMARY

A LIDAR system includes a light source that outputs an outgoing LIDAR signal that includes multiple different channels. The LIDAR system also generate multiple composite light signals that each carries a signal couple and are each associated with a different one of the channels. A signal couple includes a reference signal and an associated comparative signal. The comparative signals each include light from the outgoing LIDAR signal that has been reflected by one or more objects located outside of the LIDAR system. The reference signals also include light from the outgoing LIDAR signal but also exclude light that has been reflected by any object located outside of the LIDAR system. There is a frequency differential between a frequency of the reference signal and a frequency of the associated comparative signal. The frequency differential includes a contribution from a frequency offset that is induced by electronics. The electronics induce the frequency offset such that the frequency offset is different for each signal couple.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1A is a schematic of a LIDAR system.

FIG. 1B is a schematic of another embodiment of a LIDAR system.

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FIG. 2A illustrates a light source that includes multiple laser sources.

FIG. 2B illustrates the frequencies of LIDAR output signals that carry different channels as a function of time.

FIG. 2C illustrates an example of different light signals that can results from the LIDAR output signal of FIG. 2B.

FIG. 3A through FIG. 3C illustrate an example of suitable processing component for use in the above LIDAR systems. FIG. 3A is a schematic of the processing unit.

FIG. 3B illustrates the relationship between the frequencies associated with different channels in a data signal processed by the schematic of FIG. 3A.

FIG. 3C illustrates a schematic for an example of electronics that are suitable for use with a processing unit constructed according to FIG. 3A.

FIG. 3D illustrates a schematic for another example of electronics that are suitable for use with a processing unit constructed according to FIG. 3A.

FIG. 4A through FIG. 4C illustrate an example of suitable processing component for use in the above LIDAR systems. FIG. 4A is a schematic of the processing unit.

FIG. 4B illustrates the relationship between the frequencies associated with different channels in a data signal processed by the schematic of FIG. 4A.

FIG. 4C illustrates a schematic for an example of electronics that are suitable for use with a processing unit constructed according to FIG. 4A.

FIG. 4D illustrates a schematic for another example of electronics that are suitable for use with a processing unit constructed according to FIG. 4A.

FIG. 5 illustrates an example of a demultiplexing component that includes beam steering capability.

DESCRIPTION

The LIDAR system concurrently generates an outgoing light signal that includes multiple channels that are each of a different wavelength. The different channels are directed to the same sample region in a field of view or to different sample regions in the field of view and LIDAR data (distance and/or radial velocity between the source of a LIDAR output signal and a reflecting object) is generated for each of the sample regions. The concurrent use of multiple different channels to generate LIDAR data accelerates the generation of LIDAR data for a field of view and accordingly allows the LIDAR specifications to be satisfied for applications that require larger fields of view, increased numbers of sample regions, further field of view distances, and lower re-fresh rates.

The LIDAR system also generate multiple composite light signals that each carries a signal couple and are each associated with a different one of the channels. A signal couple includes a reference signal and an associated comparative signal. The comparative signals each include light from the outgoing LIDAR signal that has been reflected by one or more objects located outside of the LIDAR system. The reference signals also include light from the outgoing LIDAR signal but also exclude any light that has been reflected by any object located outside of the LIDAR system. There is a frequency differential between a frequency of the reference signal and a frequency of the associated comparative signal. The frequency differential includes a contribution from a frequency offset that is induced by electronics. The electronics induce the frequency offset such that the frequency offset is different for each signal couple.

Prior efforts to create the frequency offset have used a different light source for the reference signal and the comparative signal. However, the disclosed LIDAR system creates the frequency offset using reference signals and comparative signals from the same light source. As a result, the disclosed LIDAR system reduces the number of light sources that are required and accordingly reduces the number of light sources that are needed.

FIG. 1A is a schematic of a LIDAR system. The system includes a light source **10** such as a laser that outputs an outgoing LIDAR signal. The outgoing LIDAR signal carries one or more channels. When the outgoing LIDAR signal carries multiple different channels, the different channels can each be at a different wavelength. In some instances, the wavelengths of the channels are periodically spaced in that the wavelength increase from one channel to the next channel is constant or substantially constant. A suitable light source **10** for generating multiple channels with periodically spaced wavelengths includes, but is not limited to, comb lasers, multiple single wavelength lasers multiplexed into to single optical waveguide, sources such as that described in U.S. patent application Ser. No. 11/998,846, filed on Nov. 30, 2017, granted U.S. Pat. No. 7,542,641, entitled "Multi-Channel Optical Device," and incorporated herein in its entirety.

The LIDAR system also includes a utility waveguide **12** that receives an outgoing LIDAR signal from the light source **10**. A modulator **14** is optionally positioned along the utility waveguide **12**. The modulator **14** can be configured to modulate the power of the outgoing LIDAR signal and accordingly the resulting LIDAR output signal(s). Electronics **62** can operate the modulator **14**. Accordingly, the electronics can modulate the power of the outgoing LIDAR signal and accordingly the LIDAR output signal(s). Suitable modulators **14** include, but are not limited to, PIN diode carrier injection devices, Mach-Zehnder modulator devices, and electro-absorption modulator devices. When the modulator **14** is constructed on a silicon-on-insulator platform, a suitable modulator is disclosed in U.S. Pat. No. 6,177,810, filed on Sep. 21 1993, entitled Integrated Silicon PIN Diode Electro-Optic Waveguide, and incorporated herein in its entirety.

An amplifier **16** is optionally positioned along the utility waveguide **12**. Since the power of the outgoing LIDAR signal is distributed among multiple channels, the amplifier **16** may be desirable to provide each of the channels with the desired power level on the utility waveguide **12**. Suitable amplifiers include, but are not limited to, semiconductor optical amplifiers (SOAs).

The utility waveguide **12** carries the outgoing LIDAR signal from the modulator **14** to a signal-directing component **18**. The signal-directing component **18** can direct the outgoing LIDAR signal to a LIDAR branch **20** and/or a data branch **22**. The LIDAR branch outputs one or more LIDAR output signals from the LIDAR chip and receives LIDAR input signals that result from reflection of the one or more LIDAR output signals. The data branch processes the LIDAR input signals for the generation of LIDAR data (distance and/or radial velocity between the source of the LIDAR output signal and a reflecting object).

The LIDAR branch includes a LIDAR signal waveguide **24** that receives at least a portion of the outgoing LIDAR signal from the signal-directing component **18**. An output component **26** receives the outgoing LIDAR signal from the LIDAR signal waveguide **24** and outputs the outgoing LIDAR signal such that the outgoing LIDAR signal exits from the LIDAR chip. The outgoing LIDAR signal exit from

the LIDAR chip as one or more LIDAR output signals that travel through the atmosphere in which the LIDAR system is positioned. Each of the LIDAR output signals carries one of the channels carried in the outgoing LIDAR signal.

The LIDAR output signals can be reflected by a reflecting object (not shown) located outside of the LIDAR system. Each of the reflected LIDAR output signals travel through the atmosphere in which the LIDAR system is positioned and returns to the output component **26** as a LIDAR input signal. The output component **26** receives the LIDAR input signals and outputs the result on the LIDAR signal waveguide **24** as an incoming LIDAR signal.

When the outgoing LIDAR signal includes multiple different channels at different wavelengths, the output component **26** can be configured such that the LIDAR output signals carrying different channels are incident on the same sample region in the field of view or such that the LIDAR output signals carrying different channels are incident on different sample regions in the field of view. For instance, the output component **26** can be configured such that LIDAR output signals carrying different channel travels away from the LIDAR chip in different directions or such that LIDAR output signals carrying different channels travel away from the LIDAR chip in the same direction or in substantially in the same direction and at least partially overlap one another at the maximum distance at which the LIDAR system is configured to generate LIDAR data.

In some instances, the output component **26** also includes beam steering functionality. In these instances, the output component **26** can be in electrical communication with electronics **62** that can operate the output component **26** so as to steer the LIDAR output signals from one of the sample regions in a field of view to other sample regions in the field of view. The output component **26** and/or electronics can be configured such that the different LIDAR output signals are steered concurrently or independently.

Although the output component **26** is illustrated as a single component, the output component **26** can include multiple optical components and/or electrical components. Suitable output components **26** include, but are not limited to, optical phased arrays (OPAs), transmission diffraction gratings, reflection diffraction gratings, and Diffractive Optical Elements (DOE). Suitable output components **26** with beam steering capability include, but are not limited to, optical phased arrays (OPAs) with active phase control elements on the array waveguides.

The LIDAR signal waveguide **24** carries the incoming LIDAR signal to the signal-directing component **18**. The signal-directing component **18** directs the incoming LIDAR signal to the utility waveguide **12** and/or a comparative signal waveguide **28**. The portion of the incoming LIDAR signal-directed to the comparative signal waveguide **28** serves a comparative signal. The comparative signal waveguide **28** carries the comparative signal to the processing component **34**.

The signal-directing component **18** is configured such that when the signal-directing component **18** directs at least a portion of the incoming LIDAR signal to the comparative signal waveguide **28**, the signal-directing component **18** also directs at least a portion of the outgoing LIDAR signal to a reference signal waveguide **36**. The portion of the outgoing LIDAR signal received by the reference signal waveguide **36** serves as a reference signal. The reference signal waveguide **36** carries the reference signal to the processing component **34**.

As will be described in more detail below, the processing component **34** combines the comparative signal with the

reference signal to form a composite signal that carries LIDAR data for one or more sample regions in the field of view. Accordingly, the composite signal can be processed so as to extract LIDAR data for one or more sample regions in the field of view.

The signal-directing component **18** can be an optical coupler. When the signal-directing component **18** is an optical coupler, the signal-directing component **18** directs a first portion of the outgoing LIDAR signal to the LIDAR signal waveguide **24** and a second portion of the outgoing LIDAR signal to the reference signal waveguide **36** and also directs a first portion of the incoming LIDAR signal to the utility waveguide **12** and a second portion of the incoming LIDAR signal to the comparative signal waveguide **28**. Accordingly, the second portion of the incoming LIDAR signal can serve as the comparative signal and the second portion of the outgoing LIDAR signal can serve as the reference.

The signal-directing component **18** can be an optical switch such as a cross-over switch. A suitable cross-over switch can be operated in a cross mode or a pass mode. In the pass mode, the outgoing LIDAR signal is directed to the LIDAR signal waveguide **24** and an incoming LIDAR signal would be directed to the utility waveguide **12**. In the cross mode, the outgoing LIDAR signal is directed to the reference signal waveguide **36** and the incoming LIDAR signal is directed to the comparative signal waveguide **28**. Accordingly, the incoming LIDAR signal or a portion of the incoming LIDAR signal can serve as the comparative light signal and the outgoing LIDAR signal or a portion of the outgoing LIDAR signal can serve as the reference light signal.

An optical switch such as a cross-over switch can be controlled by the electronics. For instance, the electronics can control operate the switch such that the switch is in the cross mode or a pass mode. When LIDAR output signals are to be transmitted from the LIDAR system, the electronics operate the switch such that the switch is in the pass mode. When LIDAR input signals are to be received by the LIDAR system, the electronics operate the switch such that the switch is in the cross-over mode. In an embodiment disclosed below, the electronics operate the switch in the pass mode during an output period and operate the switch in the crossover mode during the feedback period. As a result, the composite signals and/or the reference signals are not generated during at least a portion of the output period and/or the LIDAR output signals are not generated during at least a portion of the feedback period. The use of a switch can provide lower levels of optical loss than are associated with the use of an optical coupler as the signal-directing component **18**.

In the above descriptions of the operation of the signal-directing component **18**, the comparative light signals and the reference light signals are concurrently directed to the data branch. As a result, the processing component **34** can combine the comparative signal with the reference signal.

In some instances, an optical amplifier **42** is optionally positioned along the LIDAR signal waveguide **24** and is configured to provide amplification of the outgoing LIDAR signal and/or of the incoming LIDAR signal. Accordingly, the effects of optical loss at the signal-directing component **18** can be reduced.

FIG. 1B illustrates the LIDAR system of FIG. 1A modified to include an optical circulator as the signal-directing component **18**. The optical circulator is configured such that the outgoing LIDAR signal is directed to the LIDAR signal waveguide **24** and the incoming LIDAR signal is directed to

the comparative signal waveguide **28**. The comparative signal waveguide **28** carries the comparative signal to the processing component **34**. Additionally, a tap component **44** is positioned along the utility waveguide **12**. The tap component **44** is configured to tap off a first portion of the outgoing LIDAR signal such that the first portion of the outgoing LIDAR signal is received on the reference signal waveguide **36**. The first portion of the outgoing LIDAR signal received by the reference signal waveguide **36** serves as the reference signal. The reference signal waveguide **36** carries the reference signal to the processing component **34**. Accordingly, the electronics can operate the LIDAR system of FIG. 1B as disclosed in the context of FIG. 1A. Suitable optical circulators include, but are not limited to, Faraday rotator based optical fiber circulators, and integrated optical circulators. Although the signal-directing component **18** of FIG. 1B is disclosed as an optical circulator, the signal-directing component **18** of FIG. 1B can be an optical coupler or optical switch.

As noted above, one or more of the light sources **10** can be a comb laser. However, other constructions of the light source **10** are possible. For instance, FIG. 2A illustrates an example of a light source **10** that includes multiple laser sources **84**. In some instances, each of the laser sources **84** outputs one or more of the channels on a source waveguide **86**. The source waveguides **86** carry the channels to a laser multiplexer **88** that combines the channels so as to form a light signal that is received on a channel waveguide or the utility waveguide **12**. Suitable laser multiplexers **88** include, but are not limited to, Arrayed Waveguide Grating (AWG) multiplexers, echelle grating multiplexers, and star couplers. The electronics can operate the laser sources **84** so the laser sources **84** concurrently output each of the channels. The electronics can operate the laser sources **84** so the laser sources **84** concurrently output each of the channels.

In some instances, each of the laser sources **84** outputs one of the channels on a source waveguide **86**. The total number of laser sources **84** included in the light source **10** can be greater than or equal to the number of LIDAR output signals that are concurrently directed to a sample region. In some instances, total number of laser sources **84** included in the light source **10** is equal to the number of LIDAR output signals that are concurrently directed to a sample region. As a result, each laser sources **84** can be the source of a different one of the LIDAR output signals that are concurrently directed to a sample region.

The electronics can operate the laser sources **84** independently. For instance, the electronics can operate the laser sources **84** so as to provide the LIDAR output signals with a particular frequency versus time waveform. Since the electronics can operate the laser sources **84** independently and each laser sources **84** can be the source of a different one of the LIDAR output signals, the electronics can operate the laser sources **84** so different LIDAR output signals have different frequency versus time waveforms.

A modulator **14** can optionally be positioned along one or more of the source waveguides **86**. The modulator **14** can each be configured to modulate the power of one of the channels and accordingly the amplitude of the resulting LIDAR output signal(s). The electronics can operate the modulator **14**. Accordingly, the electronics can modulate the power of the LIDAR output signal(s). Suitable modulators **14** include, but are not limited to, PIN diode carrier injection devices, Mach-Zehnder modulator devices, and electro-absorption modulator devices. When the modulator **14** is constructed on a silicon-on-insulator platform, a suitable modulator is disclosed in U.S. Pat. No. 617,810, filed on

Sep. 21 1993, entitled Integrated Silicon PIN Diode Electro-Optic Waveguide, and incorporated herein in its entirety.

The electronics can operate the modulators and/or the laser sources **84** so as to provide different LIDAR output signals with different waveforms. For instance, the electronics can operate one or more laser sources **84** to each produce a LIDAR output signal with a frequency that is not a function of time and an amplitude that is not a function of time such as a continuous wave. Additionally or alternately, the electronics can operate one or more laser sources **84** and associated modulator(s) **14** so as to generate one or more LIDAR output signals that has an amplitude that is a function of time. Additionally or alternately, the electronics can operate one or more laser sources **84** and associated modulator(s) **14** so as to generate a LIDAR output signal with a frequency that is a function of time. Additionally or alternately, the electronics can operate one or more laser sources **84** and associated modulator(s) **14** so as to generate a LIDAR output signal with a frequency that is a function of time and an amplitude that is a function of time.

During operation of the LIDAR system, the generation of LIDAR data is divided into a series of cycles where LIDAR data is generated for each cycle. Each LIDAR data result can be associated with a sample region in the field of view in that the resulting LIDAR data is the LIDAR data for one or more object located in that sample region. For instance, when multiple LIDAR output signals are directed to the same sample region in a field of view, one or more LIDAR data results can be generated from each one of all or a portion of the multiple LIDAR output signals during that cycle and each of the LIDAR data results can be LIDAR data for an object in that sample region. When multiple LIDAR output signals are directed to different sample regions in a field of view, one or more LIDAR data results can be generated from each one of all or a portion of the multiple LIDAR output signals during that cycle and all or a portion of the LIDAR data results can be for an object located in a different sample region in the field of view. In some instances, the one or more LIDAR output signals are directed to different sample regions in different cycles. As a result, the LIDAR data generated during different cycles is generated for different selections of the sample regions until generation of the LIDAR data for the field of view is complete and the LIDAR system repeats the process of generating LIDAR data for the field of view. When the LIDAR system repeatedly generates LIDAR data for the field of view, the LIDAR system can return the one or more LIDAR output signals to the same sample regions for which LIDAR data was previously generated.

The cycles can be performed such that the duration of each cycle can be divided into different time periods. For instance, the duration of a cycle can include one or more data periods where the LIDAR input signals are generated and received at the LIDAR chip and one or more re-location periods where the LIDAR output signal is moved from one sample region to another sample region. In a continuous scan mode, the cycle does not include any re-location periods and the LIDAR output signal is moved continuously. In one example, the cycles include multiple data periods and multiple different LIDAR output signals each carries a different channel to the same sample region. In another example, the cycles include multiple data periods and multiple different LIDAR output signals each carries a different channel to the same sample region.

FIG. 2B shows an example of a relationship between the frequency of multiple different LIDAR output signals, time, cycles and the data periods. The LIDAR output signals are

each associated with a channel index i that starts at 0 and goes to N where $N+1$ is the number of LIDAR output signals. Different LIDAR output signals are labeled λ_i in FIG. 2B. The base frequency of each LIDAR output signal (bf_i) can be the lowest frequency of the LIDAR output signal during a cycle and can be: $bf_i = f_o + i * \Delta f$. Accordingly, the base frequencies can be linearly spaced. Although FIG. 2B shows the frequencies of three LIDAR output signals labeled λ_0 , λ_1 and λ_2 ; the LIDAR system can output only one LIDAR output signal or more than three output signals.

FIG. 2B shows frequency versus time for a sequence of two cycles labeled cycle _{j} and cycle _{$j+1$} . In some instances, the frequency versus time pattern is repeated in each cycle as shown in FIG. 2B. The illustrated cycles do not include re-location periods and/or re-location periods are not located between cycles. As a result, FIG. 2B illustrates the results for a continuous scan.

Each cycle includes K data periods that are each associated with a period index k and are labeled DP_k . In the example of FIG. 2B, each cycle includes three data periods labeled DP_k with $k=1, 2$, and 3 . In some instances, the frequency versus time pattern is the same for the data periods that correspond to each other in different cycles as is shown in FIG. 2B. Corresponding data periods are data periods with the same period index. As a result, each data period DP_1 can be considered corresponding data periods and the associated frequency versus time patterns are the same in FIG. 2B. At the end of a cycle, the electronics return the frequency of each channel to the same frequency level. For instance, in FIG. 2B, the electronics return the frequency of channel i to $f_o + i * \Delta f$ to start each cycle.

Each data period includes an output period labeled OP and a feedback period labeled FP. During the output period and the feedback period, the frequency changes at a linear rate α . The rate and direction of change α is the same during the output period and the feedback period in the same data period. However, the rate of change can be different for different channels and/or for different data periods.

For a portion of the channels or for all of the channels, there is an offset period (labeled FSP) between the output periods and the feedback periods included in all or a portion of the data periods included in each cycle. During the offset period, the frequency of the LIDAR output signal and the associated reference signal changes by an amount of the frequency offset labeled fs_i . In FIG. 2B, the channel λ_0 does not have a frequency offset between the output periods and the feedback periods while the channels λ_1 has a frequency offset fs_1 between the output periods and the feedback periods and λ_2 have frequency offset fs_2 between the output periods and the feedback periods.

The frequency offsets for different channels can be different. For instance, the change in the frequency offset between channels that are adjacent to one another on the wavelength spectrum can be separated by a constant. In one example, $fs_i = W + i * (df)$ where W is a constant that can be zero or non-zero and df is a constant that represents the change in the frequency offset for channels that are adjacent to one another on the wavelength spectrum. The value of df can be positive or negative. In FIG. 2B, df is shown as positive. In one example, the frequency offsets for different channels are different and one of the frequency offsets is zero for one of the channels. In another example, the frequency offsets for different channels are different and none of the channels has a frequency offset that is zero.

FIG. 2C is a graph of frequency versus time showing the relationship between a LIDAR output signal and the resulting LIDAR input signal during the data period labeled DP_1 .

During the output period, the frequency of the LIDAR output signal can be represented by Equation 1: $f_i = f_o + i * \Delta f + \alpha * t$ where α represents that rate of frequency change and t represents time and is equal to zero at the start of the output period. During the feedback period, the frequency of the LIDAR output signal can be represented by Equation 2: $f_i = f_o + i * \Delta f + \alpha * OP + f_s + \alpha * t$ where α represents the rate of frequency change, OP represents the duration of the output period and t represents a time that is equal to zero at the start of the feedback period.

FIG. 2C also shows an example of the LIDAR input signals that can result from the illustrated LIDAR output signal. As described above, the LIDAR output signal travels away from the LIDAR chip and is reflected by an object located off of the LIDAR system. The reflected LIDAR output signal returns to the LIDAR system as the LIDAR input signal in FIG. 2C. The roundtrip time between the output of the LIDAR output signal and the receipt of the LIDAR input signal is labeled τ in FIG. 2C. The maximum roundtrip time for which the LIDAR system is configured to generate reliable LIDAR data is labeled τ_{max} in FIG. 2C. Although the LIDAR input signal shown in FIG. 2C is shown as returning to the LIDAR system at the time labeled τ , the LIDAR input signal can return to the LIDAR system with any roundtrip time greater than or equal to zero and up to τ_{max} .

As will become evident below, the LIDAR data is generated from the one or more LIDAR input signals and the one or more reference signals that occur during the feedback period. As a result, the LIDAR system can be constructed such that $\tau \leq \tau_{max} \leq OP$. In this configuration, any LIDAR output signals that experience the longest desirable roundtrip time (τ_{max}) will begin returning to the LIDAR system before or at the start of the feedback period. As will become evident below, the LIDAR data is generated from light signals that occur during the feedback period. Since $\tau_{max} \leq OP$, LIDAR input signals with a roundtrip time of τ_{max} will be returning to the LIDAR system during the feedback period. As a result, LIDAR data can be generated for these signals. In some instances, $\tau_{max} = OP$ as shown in FIG. 2B.

Additionally or alternately, the LIDAR system can be configured to have a FP selected such that $FP + FSP \leq \tau$. In some instances, FSP is zero, essentially zero, or is small enough relative to FP that FSP can be treated as zero. As a result, in some instances, FP is selected such that $FP \leq \tau$. In these configurations, any LIDAR output signals output during the feedback period, i.e. after the output period, are not received or are not substantially received as LIDAR input signals during the feedback period. As a result, the LIDAR output signals output after the frequency offset do not return to the LIDAR system during the feedback period and accordingly do not interfere with the LIDAR input signal(s) that occur during the feedback period. In some instances, the LIDAR system is configured such that $FP + FSP \leq \tau_{max}$ or $FP \leq \tau_{max}$. For instance, the LIDAR system can be used in applications where the values of τ are such that it is generally $FP \leq \tau_{max}$ or $FP + FSP \leq \tau_{max}$.

The LIDAR data is generated from the comparative signals and the reference signals that occur during the feedback period but not during the output period. For instance, the light from the portion of the LIDAR output signal(s) that are output during the output period becomes the comparative incoming LIDAR signal during the feedback period and accordingly becomes the comparative light signals during the feedback period. Since the LIDAR data is generated from the comparative light signals that occur during the feedback period, the light from this portion of the

LIDAR output signal(s) is used in the generation of the LIDAR data. In contrast, the light from the portion of the LIDAR output signal(s) that are output during the feedback period are not used in the generation of the LIDAR data.

As discussed above, the light in the LIDAR output signal(s) comes from an outgoing LIDAR signal that is also the source of the light for the reference signals. The reference signals that occur during the feedback period are used in the generation of the LIDAR data. However, the reference signals that are generated during the output period are not used in the generation of the LIDAR data. This result can be achieved by the electronics using comparative and reference signals generated during the feedback period to generate the LIDAR data but not using comparative and reference signals generated during the output period to generate the LIDAR data.

Accordingly, the light in the outgoing LIDAR signal that becomes the LIDAR output signal(s) during the output portion is used in the generation of the LIDAR data; however, during the feedback portion, the light in the outgoing LIDAR signal that becomes the LIDAR output signal(s) is not used in the generation of the LIDAR data. Additionally, the light in the outgoing LIDAR signal that becomes the reference signal(s) during the output portion is not used in the generation of the LIDAR data; however, the light in the outgoing LIDAR signal that becomes the reference signal(s) during the feedback portion is used in the generation of the LIDAR data.

Since the LIDAR output signals and the reference signals both come from the same outgoing LIDAR signal, the reference signals have the same or about the same frequency of the LIDAR output signal(s) at the same point in time. As a result, the portion of the LIDAR output signal in FIG. 2C that occurs during the feedback period is labeled reference signal. Additionally, since the light for the comparative signals comes from the LIDAR input signals, the comparative signals have the same or about the same frequency of the LIDAR input signal(s) at the same point in time. As a result, the portion of the LIDAR input signal in FIG. 2C that occurs during the feedback period is labeled comparative signal. The portion of FIG. 2C labeled reference signals and comparative signals show the frequency versus time of the comparative and reference signals that are combined to form the composite signals disclosed above. Additionally, since the labeled reference signals and comparative signals occur during the feedback period, these represent the components of the composite signal from which the LIDAR data is generated. The reference signal and the comparative signal for each channel originates from a common light source and/or a common laser sources. As a result, the number of lasers needed to produce channels with reference signals and comparative signals with frequency differentials that have different frequency offsets is reduced.

The frequency axis of FIG. 2C includes several equations that show the frequency at multiple different locations on the axis. Several of the equations are particular to one of the curves shown on the graph. For instance, the equation shown after "RS:" provides the frequency for the reference signal. The equation shown after "CS:" provides the frequency for the comparative signal. The equation shown after "LOS:" provides the frequency for the LIDAR output signal. The frequency differential between the reference signals can be determined by subtracting the equation shown after "CS:" from the equation shown after "RS:" to get $\alpha * \tau + f_s$. This result shows that the frequency differential has at least two sources. For instance, a portion of this frequency differential is the frequency offset f_s , and another portion of this fre-

quency differential is a result of the roundtrip delay, τ . The frequency offset is electronically encoded into the reference signals and the comparative signals by the electronics through use of the one or more modulators described above and/or operation of the light source **10**. In contrast, the frequency differential resulting from roundtrip delay is induced by the distance between the LIDAR system and the object off which the LIDAR output signals are reflected.

Although not shown in FIG. 2C, an example of the frequency of the LIDAR output signal during the output period for the data period labeled DP₂ in FIG. 2B can be represented by Equation 3: $f_i = f_o + i \cdot \Delta f + f_s + \alpha \cdot (OP + FP - t)$ where OP represents the duration of the output period, FP represents the duration of the feedback period, and t represents a time that is equal to zero at the start of the output period. An example of the frequency of the LIDAR output signal during the feedback period for the data period labeled DP₂ in FIG. 2B can be represented by Equation 4: $f_i = f_o + i \cdot \Delta f + 2f_s + \alpha \cdot (FP - t')$ where OP represents the duration of the output period and t' represents time and is equal to zero at the start of the feedback period.

In FIG. 2B, the rate of frequency change (α) is shown as being the same in data period DP₁ and in data period DP₂. However, the rate of frequency change (α) can be different in data period DP₁ and data period DP₂. Additionally or alternately, although the rate of frequency change (α) is shown as being the same for corresponding data periods in different channels, the rate of frequency change (α) can be different in corresponding data periods from different channels.

The data period labeled DP₃ in FIG. 2B is optional. The frequency of the LIDAR output signal during the output period for the data period labeled DP₃ in FIG. 2B can be a constant. The frequency of the LIDAR output signal during the feedback period for data period DP₃ can also be a constant where the difference between the frequency during the feedback period and during the output period is equal to f_s . Although the frequency of the LIDAR output signal during data period DP₃ is shown as a constant, the frequency can also change at a rate α . When the data period DP₃ is used to identify corresponding frequencies as described below, the rate of frequency change during data period DP₃ can be different from the rates of frequency change during data period DP₁ and data period DP₂. Although FIG. 2B and FIG. 2C are disclosed using examples where the cycles have two data periods or three data periods, the cycles can one data period or more than three data periods.

The outgoing LIDAR signal and/or the channels can be modulated so as to produce a modulated outgoing LIDAR signal and accordingly, a LIDAR output signal that is a function of a sinusoid with a frequency provided by the above frequencies f_i . As an example, the outgoing LIDAR signal and/or the channels can be modulated so as to produce a LIDAR output signal with an electrical field magnitude that is a function of or is represented by the following Equation 5: $N + M \cdot \cos(f_i \cdot t + D)$ where t can represent the t or t' defined above and M , N and, D are constants where N and D can be zero or non-zero and M is not equal to zero.

One example of a LIDAR system includes a light source constructed according to FIG. 2A where the light source is configured to generate two LIDAR output signals. One of the LIDAR output signals carries a channel with a frequency versus time according to channel λ_o of FIG. 2B and the other LIDAR output signal carries a channel with a frequency versus time according to channel λ_1 of FIG. 2B. Accordingly, the LIDAR system can be constructed according to FIG. 1A with two processing components **34**. Another

example of a LIDAR system includes a light source constructed according to FIG. 2A where the light source is configured to generate three LIDAR output signals. One of the LIDAR output signals carries a channel with a frequency versus time according to channel λ_o of FIG. 2B, another LIDAR output signal carries a channel with a frequency versus time according to channel λ_1 of FIG. 2B, and another LIDAR output signal carries a channel with a frequency versus time according to channel λ_2 of FIG. 2B. Accordingly, the LIDAR system can be constructed according to FIG. 2A with three processing components **34**. As is evident from these examples, the number of processing components **34** included in the LIDAR system can match the number of LIDAR output signals that each carries a different channel.

Suitable laser sources **84** for use with a light source **10** constructed according to FIG. 2A include, but are not limited to, external cavity lasers, distributed feedback lasers (DFBs), and Fabry-Perot (FP) lasers. External cavities lasers are advantageous in this embodiment because of their generally narrower linewidths, which can reduce noise in the detected signal.

The duration of the offset period can be short in order to increase the possible durations of the feedback period and/or the output period. For instance, the duration of the offset period can be 0.0 second and the frequency offset can accordingly be a step function. The duration of the offset period (FSP) can be non-zero as shown in FIG. 2C. In order to illustrate a non-zero duration for the offset period (FSP), FIG. 2C shows an offset period (FSP) duration that may be considered exaggerated relative to the output period (OP) and the feedback period (FP) for some embodiments of the LIDAR system.

In some instances, the duration of offset period (FSP) is greater than or equal to 0.0%, 0.05%, or 1% of the duration of the output period (OP) and/or less than 2%, 5%, or 10% of the duration of the output period (OP) and/or the duration of offset period (FSP) is greater than or equal to 0.0%, 0.1%, or 0.2% of the duration of the feedback period (FP) and/or less than 0.5%, 1%, or 2% of the duration of the feedback period (FP). Additionally or alternately, in some instances, the offset period (FSP) duration is greater than or equal to 0.0 μ s, 0.01 μ s, or 0.05 μ s and/or less than 0.1 μ s, 0.5 μ s, 1 μ s and/or the frequency change rate during the duration of the offset period (FSP) is greater than or equal to 0.1 GHz/ μ s, 0.5 GHz/ μ s, or 1 GHz/ μ s and/or less than 10 GHz/ μ s, 100 GHz/ μ s. The value of these variables can be application specific and many applications can use or require one or more variable values that are outside of the given ranges.

FIG. 3A through FIG. 3B illustrate an example of suitable processing components **34** for use in the LIDAR system of FIG. 1A and FIG. 1B. The processing unit includes a first splitter **102** that divides a reference signal carried on the reference signal waveguide **36** onto a first reference waveguide **110** and a second reference waveguide **108**. The first reference waveguide **110** carries a first portion of the reference signal to a light-combining component **111**. The second reference waveguide **108** carries a second portion of the reference signal to a second light-combining component **112**.

The processing unit includes a second splitter **100** that divides the comparative signal carried on the comparative signal waveguide **28** onto a first comparative waveguide **104** and a second comparative waveguide **106**. The first comparative waveguide **104** carries a first portion of the comparative signal to the light-combining component **111**. The

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second comparative waveguide **108** carries a second portion of the comparative signal to the second light-combining component **112**.

The second light-combining component **112** combines the second portion of the comparative signal and the second portion of the reference signal into a second composite signal. Due to the difference in frequencies between the second portion of the comparative signal and the second portion of the reference signal, the second composite signal is beating between the second portion of the comparative signal and the second portion of the reference signal. The light-combining component **112** also splits the resulting second composite signal onto a first auxiliary detector waveguide **114** and a second auxiliary detector waveguide **116**.

The first auxiliary detector waveguide **114** carries a first portion of the second composite signal to a first auxiliary light sensor **118** that converts the first portion of the second composite signal to a first auxiliary electrical signal. The second auxiliary detector waveguide **116** carries a second portion of the second composite signal to a second auxiliary light sensor **120** that converts the second portion of the second composite signal to a second auxiliary electrical signal. Examples of suitable light sensors include germanium photodiodes (PDs), and avalanche photodiodes (APDs).

The first light-combining component **111** combines the first portion of the comparative signal and the first portion of the reference signal into a first composite signal. Due to the difference in frequencies between the first portion of the comparative signal and the first portion of the reference signal, the first composite signal is beating between the first portion of the comparative signal and the first portion of the reference signal. The light-combining component **111** also splits the first composite signal onto a first detector waveguide **121** and a second detector waveguide **122**.

The first detector waveguide **121** carries a first portion of the first composite signal to a first light sensor **123** that converts the first portion of the second composite signal to a first electrical signal. The second detector waveguide **122** carries a second portion of the second composite signal to a second auxiliary light sensor **124** that converts the second portion of the second composite signal to a second electrical signal. Examples of suitable light sensors include germanium photodiodes (PDs), and avalanche photodiodes (APDs).

The first reference waveguide **110** and the second reference waveguide **108** are constructed to provide a phase shift between the first portion of the reference signal and the second portion of the reference signal. For instance, the first reference waveguide **110** and the second reference waveguide **108** can be constructed so as to provide a 90 degree phase shift between the first portion of the reference signal and the second portion of the reference signal. As an example, one reference signal portion can be an in-phase component and the other a quadrature component. Accordingly, one of the reference signal portions can be a sinusoidal function and the other reference signal portion can be a cosine function. In one example, the first reference waveguide **110** and the second reference waveguide **108** are constructed such that the first reference signal portion is a cosine function and the second reference signal portion is a sine function. Accordingly, the portion of the reference signal in the second composite signal is phase shifted relative to the portion of the reference signal in the first composite signal, however, the portion of the comparative

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signal in the first composite signal is not phase shifted relative to the portion of the comparative signal in the second composite signal.

The first light sensor **123** and the second light sensor **124** can be connected as a balanced detector and the first auxiliary light sensor **118** and the second auxiliary light sensor **120** can also be connected as a balanced detector. For instance, FIG. 3B provides a schematic of the relationship between the electronics, the first light sensor **123**, the second light sensor **124**, the first auxiliary light sensor **118**, and the second auxiliary light sensor **120**. The symbol for a photodiode is used to represent the first light sensor **123**, the second light sensor **124**, the first auxiliary light sensor **118**, and the second auxiliary light sensor **120** but one or more of these sensors can have other constructions. In some instances, all of the components illustrated in the schematic of FIG. 3B are included on the LIDAR chip. In some instances, the components illustrated in the schematic of FIG. 3B are distributed between the LIDAR chip and electronics located off of the LIDAR chip.

The electronics connect the first light sensor **123** and the second light sensor **124** as a first balanced detector **125** and the first auxiliary light sensor **118** and the second auxiliary light sensor **120** as a second balanced detector **126**. In particular, the first light sensor **123** and the second light sensor **124** are connected in series. Additionally, the first auxiliary light sensor **118** and the second auxiliary light sensor **120** are connected in series. The serial connection in the first balanced detector is in communication with a first data line **128** that carries the output from the first balanced detector as a first data signal. The serial connection in the second balanced detector is in communication with a second data line **132** that carries the output from the second balanced detector as a second data signal. The first data signal is an electrical representation of the first composite signal and the second data signal is an electrical representation of the second composite signal. Accordingly, the first data signal includes a contribution from a first waveform and a second waveform and the second data signal is a composite of the first waveform and the second waveform. The portion of the first waveform in the first data signal is phase-shifted relative to the portion of the first waveform in the first data signal but the portion of the second waveform in the first data signal being in-phase relative to the portion of the second waveform in the first data signal. For instance, the second data signal includes a portion of the reference signal that is phase shifted relative to a different portion of the reference signal that is included in the first data signal. Additionally, the second data signal includes a portion of the comparative signal that is in-phase with a different portion of the comparative signal that is included in the first data signal. The first data signal and the second data signal are beating as a result of the beating between the comparative signal and the reference signal, i.e. the beating in the first composite signal and in the second composite signal.

As a result of the above channel configurations, the first data signal and the second data signal include unwanted signal components in addition to desired beat signals. However, the values of Δf and df can be selected such that $\Delta f > (4N+1) * df/2$ where $N+1$ represents the number of channels for which LIDAR data is to be generated. When $\Delta f > (4N+1)df/2$, the unwanted signal components the first data signal and the second data signal have a frequency above the frequency of the desired beat signals. As a result, filtering can separate the unwanted signal components from the desired beat signals.

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FIG. 3B illustrates the relationship between the channels and the frequencies in the first data signal during the feedback period. The desired beat signals associated with different wavelengths appear in separate channels in the frequency domain. The channel associated with wavelength i is centered at the frequency $i(df)$. The maximum frequency for channel i is given by $i(df)+f_{max, i}$. Accordingly, the maximum frequency for the channel with the highest frequency (Channel $i=N$) is equal to $(N)(df)+f_{max, N}$. To avoid inter-channel interference, the value of df can be selected such that $df > 2*f_{max, N}$.

The value of $f_{max, i}$ can be determined from $f_{max, i} = f_{dmax} + f_{rmax} = f_{dmax, i} + \tau_{max} * (\alpha_i)$ where $f_{dmax, i}$ represents the maximum frequency shift due to the Doppler effect that channel i should experience in the LIDAR input signal and can be approximated as $\sim 2 \times V_{max} \times f_c / c$ where V_{max} represents the maximum in the absolute value of radial velocity level that should be seen by the LIDAR system, c represents speed of light, f_{rmax} represents the max frequency shift due to separation between the source of the LIDAR output signal and the reflecting object, α_i represents the value of α during the data period, and τ_{max} represents the maximum delay that should occur between transmission of a LIDAR output signal and the receipt of the LIDAR input signal. The values of variables such as α_i , τ_{max} , and V_{max} can be dependent on the application of the LIDAR system. In general, suitable values include, but are not limited to, α_i greater than 0.1 GHz/ μ s and less than 10 GHz/ μ s, τ_{max} greater than 0.1 μ s and less than 2 μ s, and V_{max} greater than 0.0 m/s and less than 150 m/s. In an example of a self-driving car application, suitable values include, but are not limited to, α_i greater than 0.1 GHz/ μ s and less than 5 GHz/ μ s, τ_{max} greater than 0.7 μ s and less than 1.4 μ s, and V_{max} greater than 100 m/s and less than 150 m/s. In an example of an Augmented Reality (AR) application, suitable values include, but are not limited to, α_i greater than 0.1 GHz/ μ s and less than 1 GHz/ μ s, τ_{max} greater than 15 ns and less than 80 μ s, and V_{max} greater than 0 m/s and less than 0.5 m/s. In some instances, df is greater than 1, 2, or 5 times the value of $2*f_{max, N}$.

Although FIG. 3B is disclosed as representing the frequencies in the first data signal, FIG. 3B can also represent the frequencies in the second data signal. Accordingly, the values of BW_i and $f_{max, i}$ associated with the first data signal can be the same for the second data signal.

In some instances, the LIDAR data is generated for each of the channels by providing the first data signal to a first Analog-to-Digital Converter (ADC) and the second data signal to a second Analog-to-Digital Converter (ADC). The resulting digital signals can then be provided to a transform module configured to perform a complex transform on a complex signal so as to convert the input from the time domain to the frequency domain. The first data signal can be the real component of the complex signal and the second data signal can be the imaginary component of the complex signal. The transform module can execute the attributed functions using firmware, hardware and software or a combination thereof. The transform converts the input from the time domain to the frequency domain. Accordingly, the transform module can output one or more frequencies the each corresponds to an object in the sample region illuminated by the LIDAR output signal. Each of the different frequencies is used by the electronics as the frequency of the LIDAR input signal. The electronics can use the frequencies for further processing to determine the distance and/or velocity of each of the one or more reflecting objects in the sample region.

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One issue with the use of Analog-to-Digital Converters (ADC) on the first data signal and/or on the second data signal may be that the ADC sampling rate required to generate useful results may be impractical to achieve. Another option is to separate the different channels in the first data signal and the second data signal before converting from analog to digital.

FIG. 3C illustrates a schematic for an example of electronics that are suitable for use with a processing unit constructed according to FIG. 3A. The first data line 128 carries the first data signal to $N+1$ data processing branches 140 that are each associated with one of the channels. As N becomes larger, it may be desirable to place an optional amplifier 142 along the first data line 128 to amplify to the power of the first data signal to a power level that is sufficient power for processing of the first data signal by the $N+1$ data processing branches 140. The second data line 132 carries the second data signal to the $N+1$ data processing branches 140 that are each associated with one of the channels. As N becomes larger, it may be desirable to place an optional amplifier 142 along the second data line 138 to amplify to the power of the second data signal to a power level that is sufficient power for processing of the second data signal by the $N+1$ data processing branches 140.

In the data processing branches 140 for channels $i > 0$, the first data line 128 carries the first data signal to a first multiplier 144 that multiplies the first data signal by a first multiplier signal. The first multiplier signal is selected to shift the frequencies associated with channel $i > 0$ in the first data signal to a lower frequency. In the example of FIG. 3C, the first multiplier signal is selected to shift the frequencies associated with channel $i > 0$ such that the band BW_i is centered at the zero frequency. Accordingly, the first multiplier signal can be represented by $\cos(2\pi(i)(df)t)$. The first multipliers each outputs a first-frequency shifted data signal. Suitable first multipliers include, but are not limited to, RF mixers such as a Gilbert cell mixer.

In the data processing branches 140 for channels $i > 0$, the second data line 132 carries the second data signal to a second multiplier 146 that multiplies the second data signal by a second multiplier signal. The second multiplier signal is selected to shift the frequencies associated with channels $i > 0$ in the second data signal to a lower frequency. In some instances, the frequencies associated with each channel $i > 0$ in the second data signal are shifted to the same frequency as the associated channel i in the first data signal. In the example of FIG. 3C, the second multiplier signal is selected to shift the frequencies associated with channel $i > 0$ such that the band BW_i is centered at the zero frequency. Accordingly, the second multiplier signal can be represented by $\cos((\pi/2) + 2\pi(i)(df)t)$. The second multipliers each outputs a second frequency-shifted data signal. Suitable second multipliers include, but are not limited to, RF mixers such as a Gilbert cell mixer.

The first frequency-shifted data signals and the second frequency-shifted data signals are each received on a filter input line 150 that each carries the received signal to a filter 152. In the above discussions, the frequencies associated with channels $i > 0$ are shifted so as to be centered at a frequency of 0.0. However, the frequencies associated with the channel $i=0$ in the first data signal are already centered at a frequency of 0.0. As a result, the first data line 128 can carry the first data signal to a filter 152 without the first data signal being processed by a multiplier. Additionally, the frequencies associated with the channel $i=0$ in the second data signal are already centered a frequency of 0.0. As a

result, the second data line 132 carries the second data signal to a filter 152 without the second data signal being processed by a multiplier.

The filters 152 in each data processing branches 140 are configured to pass the frequencies in a frequency band centered at zero-frequency while filtering out other frequency bands. Since the first frequency-shifted data signals in the data processing branch 140 associated with channel i have the frequency band for channel i centered at a zero frequency, each filter 152 outputs a first channel data signal that includes the LIDAR data associated with channel i but does not include or substantially include LIDAR data for the channels below channel i ($<i$) or above channel i ($>i$). Since the second frequency-shifted data signals in the data processing branch 140 associated with channel i have the frequency band for channel i centered at a zero frequency, each filter 152 outputs a second channel data signal that includes the LIDAR data associated with channel i but does not include or substantially include LIDAR data for the channels below channel i ($<i$) or above channel i ($>i$). Suitable filters for use as the first filters and/or second filters include, but are not limited to, lowpass filters because the first frequency-shifted data signals and the second frequency-shifted data signals are centered at zero-frequency.

The first channel data signals and the second channel data signals are each received on an ADC input line 154 that each carries the received signal to an Analog-to-Digital Converter 156 (ADC). According to the Nyquist sampling theorem, the sampling rate for an Analog-to-Digital Converters (ADC) is generally greater than or equal to twice the highest frequency in the signal. Accordingly, if the frequency arrangement were as shown in FIG. 3B, the sampling rate for channel i would be greater than or equal to $2 \cdot (i(df) + f_{max,i})$. In FIG. 3B, the signal i would be considered oversampled when the sampling rate is greater than $2 \cdot (i(df) + f_{max,i})$ and undersampled when sampling rate is less than $2 \cdot (i(df) + f_{max,i})$. However, because the multipliers reduced the frequency of the channels, the frequency band of the first channel data signals and the second channel data signals are not arranged as shown in FIG. 3B but have been reduced such that the highest frequency for each first channel data signal and each second channel data signal is less than or equal to $f_{max,N}$ in FIG. 3B. Accordingly, the sampling rate for each Analog-to-Digital Converter 156 can be greater than or equal to twice $2 \cdot f_{max,N}$. As a result, the down conversion provided by the multiplier reduces the sampling rate required for each Analog-to-Digital Converter 156. The Analog-to-Digital Converters 156 that each receives a first channel data signal outputs a first digital data signal. The Analog-to-Digital Converters 156 that each receives a second data signal outputs a second digital data signal. The first digital data signals and the second digital data signals are each received on a digital data line 158. Each digital data line carries the received signal to a transform module 160.

The transform modules 160 are arranged such that each transform modules 160 receives one of the first digital data signals and the associated second digital data signal. Additionally, each of the transform modules 160 is associated with a different one of the channels. The transform modules 160 associated with channel i receives the first digital data signal for channel i and also receives the second digital data signal for channel i .

The transform modules 160 are configured to perform a complex transform on a complex signal so as to convert the input from the time domain to the frequency domain. The first digital data signal can be the real component of the complex signal and the second digital data signal can be the

imaginary component of the complex signal. The transform modules can execute the attributed functions using firmware, hardware and software or a combination thereof.

The Complex Fourier transform converts the input from the time domain to the frequency domain and outputs one or more frequencies the each corresponds to an object in the sample region illuminated by the LIDAR output signal. Different reflecting objects in a sample region need not be physically separate items but can be different surfaces of the same item that are located different distances from the LIDAR system and/or are moving at different radial velocities relative to the LIDAR system as might occur with a jagged object that is both rotating and translating relative to the LIDAR system. Each of the different frequencies is used by the electronics as a frequency for the LIDAR input signal carrying channel i . The electronics can use the frequencies for further processing to determine the distance and/or velocity of each of the one or more reflecting objects in the sample region.

The electronics can use the one or more frequencies output from the transform module for channel i to determine the LIDAR data (the radial velocity and/or the distance between a reflecting object in the sample region and the source of the LIDAR output signal) for each of the one or more objects in the sample region illuminated in channel i . For instance, the following equation applies during a data period where electronics increase the frequency of the outgoing LIDAR signal during the data period such as occurs in data period DP_1 of FIG. 2B: $f_{ub} = -f_{d,i} + \alpha\tau_0$ where f_{ub} is the frequency provided by the transform module, $f_{d,i}$ represents the Doppler shift ($f_{d,i} = 2vf_{c,i}/c$) where v is the velocity of the reflecting object relative to the chip where the direction from the reflecting object toward the chip is assumed to be the positive direction, and c is the speed of light, α is the rate at which the frequency is changed during the data period, and τ_0 is the roundtrip delay between transmission of a LIDAR output signal and receipt of the corresponding LIDAR input signal. The following equation applies during a sample where electronics decrease the frequency of the outgoing LIDAR signal such as occurs in data period DP_2 of FIG. 2B: $f_{db} = -f_{d,i} - \alpha\tau_0$ where f_{db} is a frequency provided by the transform module. In these two equations, $f_{d,i}$ and τ_0 are unknowns. The electronics solve these two equations for the two unknowns. The radial velocity for the sample region then be determined from the Doppler shift ($v = c \cdot f_d / (2f_c)$) and the separation distance for that sample region can be determined from $c \cdot \tau_0 / 2$. Since the LIDAR data can be generated for each of the frequencies output by the transform, separate LIDAR data can be generated for each of the objects in a sample region. Accordingly, the electronics can determine more than one radial velocity and/or more than one radial separation distance from a single sampling of a single sample region in the field of view.

The data period labeled DP_3 in FIG. 2B is optional. As noted above, there are situations where more than one object is present in a sample region. For instance, during the feedback period in DP_1 for cycle₂ and also during the feedback period in DP_2 for cycle₂, the transform module for a channel may output more than one frequency. In these circumstances, it may not be clear which frequencies from DP_2 correspond to which frequencies from DP_1 . As a result, it may be unclear which frequencies need to be used together to generate the LIDAR data for an object in the sample region. As a result, there can be a need to identify corresponding frequencies. The identification of corresponding frequencies can be performed such that the corresponding

frequencies are frequencies from the same reflecting object within a sample region. The data period labeled DP_3 can be used to find the corresponding frequencies. LIDAR data can be generated for each pair of corresponding frequencies and is considered and/or processed as the LIDAR data for the different reflecting objects in the sample region.

An example of the identification of corresponding frequencies uses a LIDAR system where the cycles include three data periods (DP_1 , DP_2 , and DP_3) as shown in FIG. 2B. When there are two objects in a sample region illuminated by the LIDAR outputs signal for channel λ_0 , the transform module that receives the channel λ_0 outputs two different frequencies for f_{ub} , f_{u1} and f_{u2} during DP_1 and another two different frequencies for f_{db} , f_{d1} and f_{d2} during DP_2 . In this instance, the possible frequency pairings are: (f_{d1} , f_{u1}); (f_{a1} , f_{a2}); (f_{d2} , f_{u1}); and (f_{d2} , f_{du2}). A value of f_d and τ_0 can be calculated for each of the possible frequency pairings. Each pair of values for f_d and τ_0 can be substituted into $f_3 = -f_d + \alpha_3 \tau_0$ to generate a theoretical f_3 for each of the possible frequency pairings. In this case, the transform modules **136** that receives the channel λ_0 also outputs two values for f_3 that are each treated as an actual f_3 value. The frequency pair with a theoretical f_3 value closest to each of the actual f_3 values is considered a corresponding pair. LIDAR data can be generated for each of the corresponding pairs as described above and is considered and/or processed as the LIDAR data for a different one of the reflecting objects in the sample region.

Each set of corresponding frequencies can be used in the above equations to generate LIDAR data. The generated LIDAR data will be for one of the objects in the sample region. As a result, multiple different LIDAR data values can be generated for a sample region where each of the different LIDAR data values corresponds to a different one of the objects in the sample region.

FIG. 3D illustrates another example of suitable processing components for use in the above LIDAR systems. The first data line **128** carries the first data signal to $N+1$ data processing branches **140** that are each associated with one of the channels. As N becomes larger, it may be desirable to place an optional amplifier **142** along the first data line **128** to amplify to the power of the first data signal to a power level that is sufficient power for processing of the first data signal by the $N+1$ data processing branches **140**. The second data line **132** carries the second data signal to the $N+1$ data processing branches **140** that are each associated with one of the channels. As N becomes larger, it may be desirable to place an optional amplifier **142** along the second data line **138** to amplify to the power of the second data signal to a power level that is sufficient power for processing of the second data signal by the $N+1$ data processing branches **140**.

The first data line **128** carries the first data signal to a different filter **152** located in each of the data processing branches **140**. Additionally, the second data line **132** carries the second data signal to a different filter **152** located in each of the data processing branches **140**. The filters **152** in each data processing branch i are configured to pass the frequencies that fall within bandwidth i (BW_i) while filtering out frequencies that fall within bandwidths $>i$ ($BW_{>i}$) and bandwidths $<i$ ($BW_{<i}$). As a result, the filters **152** in data processing branch i that receive a first data signal output a first channel data signal that includes the LIDAR data associated with channel i but does not include or substantially include LIDAR data for the channels below channel i ($<i$) or above channel i ($>i$). Additionally, the filter **152** in data processing branch i that receive a second data signal output a second channel data signal that includes the LIDAR

data associated with channel i but not include or substantially include LIDAR data for channels below channel i ($<i$) or above channel i ($>i$).

Suitable filters **152** for use as the filter **152** in the data processing branch associated with channel $i=0$ can be a lowpass filter because the frequencies in BW_1 are centered around the zero frequency. Suitable filters **152** for use as the filter **152** in the data processing branch associated with channel $i>0$ can be bandpass filters because the frequencies in BW_i are higher than the zero frequency.

The first channel data signals and the second channel data signals are each received on an ADC input line **154** that each carries the received signal to an Analog-to-Digital Converter **156** (ADC). According to the Nyquist sampling theorem, the sampling rate for an Analog-to-Digital Converters (ADC) is generally greater than or equal to twice the highest frequency in the signal. Accordingly, in FIG. 3B, the signal i would be considered oversampled when the sampling rate is greater than $2*(i(df)+f_{max,i})$ and undersampled when sampling rate is less than $2*(i(df)+f_{max,i})$. In the system of FIG. 3D, at least a portion of the first channel data signals and the second channel data signals are undersampled. For instance, the Analog-to-Digital Converters (ADC) for the first channel data signals and the second channel data signals associated with at least channels $i>0$ are undersampled. Undersampling the first channel data signal for channel i causes an ADC to output a first digital signal that includes the frequencies of BW_i shifted to the zero-frequency. Undersampling the second channel data signal for channel i causes an ADC to output a second digital signal that includes the frequencies of BW_i shifted to the zero-frequency.

The undersampling rate can be selected based on the bandwidth of the channel $i=N$ in the first data signal. For instance, the sampling rate for each Analog-to-Digital Converter **156** can be greater than or equal to $f_{max,0}$ and less than or equal to $f_{max,N}$. As a result, the electronics of FIG. 3D can reduce the required sampling rates to the levels achieved with the down conversion provided by the electronics of FIG. 3C.

The first digital data signals and the second digital data signals are each received on a digital data line **158**. Each digital data line carries the received signal to a transform module **160**. The transform modules **160** are arranged and operated as disclosed in the FIG. 3A through FIG. 3C to generate LIDAR data for each of the channels.

In some instances, a simplified version of the processing components **34** can be employed. For instance, FIG. 4A through FIG. 4D illustrate simplified processing components **34** that are suitable for use with the above LIDAR systems. The comparative signal carried on the comparative waveguide **28** and the reference signal carried on the reference signal waveguide **36** are carried to a light-combining component **111**. The light-combining component **111** operates as disclosed above and outputs the first portion of the first composite signal and the second portion of the first composite signal as disclosed above. The remaining components operate on the first portion of the first composite signal and the second portion of the first composite signal as disclosed in the context of FIG. 3A through FIG. 3D to generate the first digital data signals that are each associated with one of the channels. The first digital data signals are each received on a digital data line **158**. Each digital data line carries the received signal to a transform module **160**.

The transform modules **160** can perform a real transform on the first digital data signals so as to convert the input from the time domain to the frequency domain. Examples of suitable transforms include a real Fourier transform that

converts the input from the time domain to the frequency domain. The transform can output multiple frequency peaks and the electronics can select which frequency peaks to be used as representing the frequency of the LIDAR input signal for comparative channel i . The electronics use the selected frequency for further processing to determine the LIDAR data. The processing components 34 disclosed in the context of FIG. 4A through FIG. 4D are most suitable for use when the source of the LIDAR signal and reflecting object have a radial velocity and radial separation within known ranges and the frequency solutions within those ranges have results that allow the correct frequency solutions to be selected.

Suitable output components 26 for use in the LIDAR system can be waveguide facets. FIG. 5 illustrates an example of a suitable output component 26 that can optionally include beam steering capability. The demultiplexing component 26 includes a splitter 184 that receives the outgoing light signal from the LIDAR signal waveguide 24. The splitter divides the outgoing light signal into multiple output signals that are each carried on a steering waveguide 186. Each of the steering waveguides ends at a facet 188. The facets are arranged such that the output signals exiting the chip through the facets combine to form the LIDAR output signal.

The splitter and steering waveguides can be constructed such that there is not a phase differential between output signals at the facet of adjacent steering waveguides. For instance, the splitter can be constructed such that each of the output signals is in-phase upon exiting from the splitter and the steering waveguides can each have the same length. Alternately, the splitter and steering waveguides can be constructed such that there is a linearly increasing phase differential between output signals at the facet of adjacent steering waveguides. For instance, the steering waveguides can be constructed such that the phase of steering waveguide number k is $f_o' + (k-1)f^*$ where k is an integer from 1 to K and represents the number associated with a steering waveguide when the steering waveguides are sequentially numbered as shown in FIG. 5, f is the phase differential between neighboring steering waveguides when the phase tuners (discussed below) do not affect the phase differential, and f_o' is the phase of the output signal at the facet of steering waveguide $k=1$. Because the channels can have different wavelengths, the values of f and f_o' can each be associated with one of the channels. In some instances, this phase differential is achieved by constructing the steering waveguides such that the steering waveguides have a linearly increasing length differential. For instance, the length of steering waveguide k can be represented by $L_o + (k-1)\Delta l$ where Δl is the length differential between neighboring steering waveguide, and L_o is the length of steering waveguide $k=1$. Because Δl is a different percent of the wavelength of different channels included in the outgoing LIDAR signal, each of the different LIDAR output signals travels away from LIDAR chip in a different direction (θ). When the steering waveguides are the same length, the value of Δl is zero and the value of f is zero. Suitable Δl include, but are not limited to, Δl greater than 0, or 5 and/or less than 10, or 15 μm . Suitable f include, but are not limited to, f greater than 0π , or 7π and/or less than 15π , or 20π . Suitable K include, but are not limited to, K greater than 10, or 500 and/or less than 1000, or 2000. Suitable splitters include, but are not limited to, star couplers, cascaded Y-junctions and cascaded 1x2 MMI couplers.

A phase tuner 190 can be positioned along at least a portion of the steering waveguides. Although a phase tuner

is shown positioned along the first and last steering waveguide, these phase tuners are optional. For instance, the chip need not include a phase tuner on steering waveguide $k=1$.

The electronics can be configured to operate the phase tuners so as to create a phase differential between the output signals at the facet of adjacent steering waveguides. The electronics can operate the phase tuners such that the phase differential is constant in that it increases linearly across the steering waveguides. For instance, electronics can operate the phase tuners such that the tuner-induced phase of steering waveguide number k is $(k-1)\omega$ where ω is the tuner-induced phase differential between neighboring steering waveguides. Accordingly, the phase of steering waveguide number k is $f_o' + (k-1)f + (k-1)\omega$. FIG. 5 illustrates the chip having only 4 steering waveguides in order to simplify the illustration, however, the chip can include more steering waveguides. For instance, the chip can include more than 4 steering waveguides, more than 100 steering waveguides, or more than 1000 steering waveguides and/or less than 10000 steering waveguides.

The electronics can be configured to operate the phase tuners so as to tune the value of the phase differential ω . Tuning the value of the phase differential ω changes the direction that the LIDAR output signal travels away from the chip (θ). Accordingly, the electronics can scan the LIDAR output signal by changing the phase differential ω . The range of angles over which the LIDAR output signal can be scanned is ϕ_R and, in some instances, extends from ϕ_v to $-\phi_v$ with $\phi=0^\circ$ being measured in the direction of the LIDAR output signal when $\omega=0$. When the value of Δl is not zero, the length differential causes diffraction such that light of different wavelengths travels away from chip in different directions (θ). Accordingly, there may be some spreading of the outgoing LIDAR signal as it travels away from the chip. Further, changing the level of diffraction changes the angle at which the outgoing LIDAR signal travels away from the chip when $\omega=0^\circ$. However, providing the steering waveguides with a length differential ($\Delta l \neq 0$) can simplify the layout of the steering waveguides on the chip.

Suitable electronics can include, but are not limited to, a controller that includes or consists of analog electrical circuits, digital electrical circuits, processors, microprocessors, digital signal processors (DSPs), Field Programmable Gate Arrays (FPGAs), computers, microcomputers, or combinations suitable for performing the operation, monitoring and control functions described above. In some instances, the controller has access to a memory that includes instructions to be executed by the controller during performance of the operation, control and monitoring functions. Although the electronics are illustrated as a single component in a single location, the electronics can include multiple different components that are independent of one another and/or placed in different locations. Additionally, as noted above, all or a portion of the disclosed electronics can be included on the chip including electronics that are integrated with the chip.

Additional details about the construction and operation of a demultiplexing component 26 constructed according to FIG. 5 can be found in U.S. Provisional Patent Application Ser. No. 62/680,787, filed on Jun. 5, 2018, and incorporated herein in its entirety.

The above disclosure uses channel assignments that start channel $i=0$ through channel N for a total of $N+1$ channels. However, the channel indices can be shifted. For instance, the channel index can be configured such that the channels start at channel $j=1$ through channel M for a total of M

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channels. Such a shift can be performed by substituting $i=j-1$ into the above equations.

Other embodiments, combinations and modifications of this invention will occur readily to those of ordinary skill in the art in view of these teachings. Therefore, this invention is to be limited only by the following claims, which include all such embodiments and modifications when viewed in conjunction with the above specification and accompanying drawings.

The invention claimed is:

1. A method of operating a LIDAR system, comprising: outputting from a light source an outgoing LIDAR signal, generating multiple composite light signals that each carries at least a signal couple, a signal couple including a reference signal and an associated comparative signal, each signal couple being associated with a different channel,

the comparative signals each including light from the outgoing LIDAR signal that has been reflected by one or more objects located outside of the LIDAR system and the reference signals each including light from the outgoing LIDAR signal but excluding light that has been reflected by any object located outside of the LIDAR system; and

outputting from the LIDAR system LIDAR output signals,

the LIDAR output signals having a frequency versus time pattern that is repeated over sequential cycles, the comparative signals each including light from a different one of the LIDAR output signals, each cycle includes multiple data periods and the frequency versus time pattern is the same for corresponding data periods in different cycles, one or more of the data periods includes an output period and a feedback period, a rate of a frequency

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change in the LIDAR output signals being the same during the output period and the feedback period of the same data period, and

an offset period being present between the output period and the feedback period in each data period that is included in the one or more data periods, the frequency of the LIDAR output signal changing by a frequency offset during the offset period.

2. The method of claim 1, wherein light included in each reference signal and in the associated comparative signal was generated by the same laser.

3. The method of claim 1, wherein each signal couple has a non-zero frequency differential.

4. The method of claim 3, wherein the frequency offset of each LIDAR output signal is a portion of the frequency differential for the signal couple having the comparative signal that includes light from the LIDAR output signal.

5. The method of claim 4, wherein a portion of the frequency differential is induced by a distance between the LIDAR system and the one or more objects located outside of the LIDAR system.

6. The method of claim 1, wherein for at least a portion of the LIDAR output signals, each cycle includes multiple data periods and each data period includes an offset period between the output period and the feedback period.

7. The method of claim 1, further comprising: generating LIDAR data from composite signals that occur during the feedback periods but not during the output periods, the LIDAR data indicating a distance and/or radial velocity between the LIDAR system and the one or more objects.

8. The method of claim 1, wherein the composite light signals are each generated by a light-combining component configured to combine one of the reference signals with the associated comparative signal.

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